MA 587 Homework 2

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Question 1 Error bound

1(a) Consider a single cell of the problem domain, $L_j = [x_{j-1}, x_j]$, and choose some $z \in (x_{j-1}, x_j)$. The quadratic polynomial that interpolates the points $\{u(x_{j-1}), u(z), u(x_j)\}$ is given by

$$\hat{u}_z(x) = u_{j-1} + \frac{u_j - u_{j-1}}{h} x + A_z(x - x_j)(x - x_{j-1}), \tag{1}$$

where A_z is some constant dependent on the choice of z, $u_j = u(x_j)$, and $h = x_j - x_{j-1}$. This can also be rewritten as

$$\hat{u}_z(x) = \tilde{u}(x) + A_z(x - x_j)(x - x_{j-1}), \tag{2}$$

where $\tilde{u}$ is the linear interpolation of u. Then the error $g(x) = \hat{u}(x) - u(x)$ is zero in at least three points: $g(x_{j-1}) = g(z) = g(x_j) = 0$. By Rolle's theorem, since g has three zeros, there are two points $\xi_{1,z} \in (x_{j-1}, z)$ and $\xi_{2,z} \in (z, x_j)$, for which $g'(\xi_{1,z}) = g'(\xi_{2,z}) = 0$. Applying Rolle's theorem once more, there is another point $\xi_z \in (\xi_{1,z}, \xi_{2,z})$ such that $g''(\xi_z) = 0$. That is,

$$g''(\xi_z) = \hat{u}''(\xi_z) - u''(\xi_z), \tag{3}$$

so, using Equation 1,

$$u''(\xi_z) = 2A_z \tag{4}$$

$$\frac{u''(\xi_z)}{2} = A_z. \tag{5}$$

This value is bounded by

$$\left|\frac{u''(\xi_z)}{2}\right| \le \max_{a\in[x_{j-1},x_j]}\left|\frac{u''(a)}{2}\right| \le \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \tag{6}$$

Then, for any $x \in [x_{j-1}, x_j]$,

$$u(x) = \tilde{u}(x) + A_x(x-x_j)(x-x_{j-1}) \tag{7}$$
$$u(x) = \tilde{u}(x) + \frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1}) \tag{8}$$
$$u(x) - \tilde{u}(x) = \frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1}), \tag{9}$$

so

$$|u(x) - \tilde{u}(x)| = \left|\frac{u''(\xi_x)}{2}(x-x_j)(x-x_{j-1})\right|. \tag{10}$$

By submultaplicativity,

$$|u(x) - \tilde{u}(x)| \le \left|\frac{u''(\xi_x)}{2}\right| \cdot |(x-x_j)(x-x_{j-1})|. \tag{11}$$

Applying the triangle inequality to the $(x-x_j)(x-x_{j-1})$ term and applying Equation 6 to the $\frac{u''}{2}$ term, this is bounded by

$$|u(x) - \tilde{u}(x)| \le \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \cdot (|x-x_j| + |x-x_{j-1}|), \tag{12}$$

or

$$|u(x) - \tilde{u}(x)| \le \max_{y\in[0,1]}\left|\frac{u''(y)}{2}\right| \cdot h \tag{13}$$
$$|u(x) - \tilde{u}(x)| \le \max_{y\in[0,1]}|u''(y)| \cdot h. \tag{14}$$

1(b) The inequality given is

$$\|(u-u_h)'\| \le h \cdot \max_{y\in[0,1]}|u''(y)|. \tag{15}$$

The Cauchy-Schwarz Inequality states that $|\langle \boldsymbol{x}, \boldsymbol{y}\rangle| \le \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$, where $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \int_0^1 \boldsymbol{x}\boldsymbol{y}dx$ is the inner product, and the norm $\|\cdot\|$ is $\langle\cdot,\cdot\rangle^{1/2}$ Let $\boldsymbol{x} = (u-u_h)'$ and let $\boldsymbol{y} = 1$:

$$\left| \langle (u - u_h)', 1 \rangle \right| \leq \|(u - u_h)'\| \cdot \|1\| \leq h \cdot \max_{y \in [0,1]} |u''(y)| \,, \tag{16}$$

and, expanding the inner product definition,

$$\left| \int_0^1 (u - u_h)' dx \right| \leq h \cdot \max_{y \in [0,1]} |u''(y)| \,. \tag{17}$$

Using a change of variables,

$$\left| \int_0^x (u - u_h)'(s) ds \right| \leq \left| \int_0^1 (u - u_h)'(s) ds \right| \leq h \cdot \max_{y \in [0,1]} |u''(y)| \tag{18}$$

for $x \in [0, 1]$, which becomes

$$|[u(x) - u_h(x)] - \cancel{[u(0) - u_h(0)]}| \leq h \cdot \max_{y \in [0,1]} |u''(y)| \tag{19}$$

$$|u(x) - u_h(x)| \leq h \cdot \max_{y \in [0,1]} |u''(y)| \,. \tag{20}$$